UNITY 6000.0.62F1 · COMPLETE OFFLINE DOCUMENTATION

Number Merge Developer Guide

Install. Edit. Reskin. Build your release.

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CHAPTER 01

Start here: Number Merge

Number Merge is a standalone editable Unity game template. Drag a sweet through empty space and push it into a sweet with the same number to merge into the next numbered sweet. Manage the rising rows and follow the visible stage goal. Sugar cubes and guided tools alter the board.

Includes 25 curated levels, source-backed generation, a game-specific visual editor and validation, 4 guided helpers, local saving and portrait/landscape scenes. Keep the required private game Kit and publisher-owned Szekipapa77Ads layer. Provider SDK binaries are excluded. There is no bundled HLK, purchases, multiplayer server or remote LiveOps backend. Import only one game Kit into a project because the Kit classes share names.

CHAPTER 02

Requirements, installation and first play

Use Unity **6000.0.62f1**, URP **17.0.4** and Unity UI **2.0.0**. Install matching Android SDK/NDK/OpenJDK and WebGL modules in Unity Hub. Retain the supplied manifest and ProjectSettings.

1. Extract the standalone project to a short writable local path, then open it in the matching editor. Wait for package import and compilation.
2. For native unitypackage import, create a matching URP project and use Assets > Import Package > Custom Package. Import every game, Kit, Szekipapa77Ads and WebGL-template entry.
3. Run Szekipapa77 > Number Merge > Setup. It configures rendering and player settings; review its changes when integrating with existing work.
4. Active Input Handling must support the classic Input Manager: choose Old or Both. New Input System only is insufficient.
5. Open `Assets/Games/NumberMerge/Scenes/NumberMerge_Mobile.unity` and press Play. Follow the level card and first-level instructions.
6. Stop Play mode before editing level assets. Keep all .meta GUIDs. The game uses classic Unity UI Text; do not assume an extra TextMesh Pro import is required.

CHAPTER 03

Project folders and exact source map

All game-relative paths below start at `Assets/Games/NumberMerge`. Runtime UI is built in code; editing an unrelated static Canvas does not reskin it. Scripts contains game rules and views; Editor contains authoring tools; Kit supplies UI/audio/save/platform support; Resources/numbermerge contains level JSON, art, music and sound; Settings owns URP assets; Scenes contains both entry scenes.

- `Scripts/NmArt.cs`
- `Scripts/NmBot.cs`
- `Scripts/NmCore.cs`
- `Scripts/NmGen.cs`
- `Scripts/NumberMergeGame.cs`
- `Scripts/NumberMergeGame.Hud.cs`
- `Scripts/NumberMergeGame.View.cs`

Main flow: NumberMergeGame.cs. UI and popup consumers: NumberMergeGame.Hud.cs. Model/generator/loading: NmCore.cs, NmGen.cs. Authoring: Editor/NumberMergeLevelEditor.cs. Shared ad settings live outside the game at Assets/Szekipapa77Ads/Resources/AdsSettings.asset.

CHAPTER 04

Gameplay, controls and progression

Drag a sweet through empty space and push it into a sweet with the same number to merge into the next numbered sweet. Manage the rising rows and follow the visible stage goal. Sugar cubes and guided tools alter the board.

The initial bank has 25 curated levels. Read the supplied loader/generator before changing continuation rules. Play the guided opening, ordinary progress, completion and retry paths; preserve saved indexes during a visual reskin. The Mobile scene is portrait and the WebGL-named scene is landscape, but the supplied Build menu selects portrait for both build targets. Select landscape explicitly in your Build Profile if desired. Test a near-startup scene replacement as well as a normal restart.

CHAPTER 05

Tutorials and every helper

Fresh local progress triggers the first-level guidance. Newly unlocked helpers have introduction and guided-use flows. Reset only a test save when reviewing tutorials. A free guided first use can differ from ordinary charge consumption. Exact helper names/unlocks/charges: Scripts/NumberMergeGame.cs; help and effect descriptions: Scripts/NumberMergeGame.Hud.cs.

Helper	Unlock	Free charges	Action and effect
Pop	3	3	Tap Pop, then tap any sweet or sugar cube to smash it.
Chill	7	2	Freezes the row timer for 15 seconds.
Sprinkle	12	2	Tap Sprinkle, then tap a sweet to add +1 to it.
Scoop	17	1	Scoops away the whole bottom row. Everything drops one step.

Activate the tool and perform its required target action: arming a tool alone does not demonstrate its effect. A successful use requires the stated preconditions from this game: show both activation and its actual effect. Exhausted charges may offer a rewarded ad; verify success/cancel/failure and grant a reward once only.

CHAPTER 06

Visual level editor: worked editing session

Open **Szekipapa77 > Number Merge > Level Editor**. Exact implementation: `Editor/NumberMergeLevelEditor.cs`.

Duplicate an early entry in this game-specific visual editor, change its visible name and one rule, save, validate, then play it. Use the exact source-control table below for this game.

1. Back up Resources/numbermerge/Levels/levels.json.
2. Duplicate a simple early level and edit one small element.
3. Use the game-specific validation controls and resolve every reported issue.
4. Save explicitly; stop and restart Play mode, then use Play this level or navigate to the edited index.
5. Check portrait and landscape and rebuild demo outputs after accepting data changes.

The exact current authoring controls, including tooltips and generator switches, follow. These are game-specific source references, not a generic editor feature list.

Source	Exact current code
`NumberMergeLevelEditor.cs:42`	`[MenuItem("Szekipapa77/Number Merge/Level Editor")]`
`NumberMergeLevelEditor.cs:45`	`[MenuItem("Szekipapa77/Number Merge/Validate All Levels")]`
`NumberMergeLevelEditor.cs:110`	`if (GUILayout.Button(Dirty ? "Save *" : "Save", EditorStyles.toolbarButton, GUILayout.Width(70))) Save();`
`NumberMergeLevelEditor.cs:111`	`if (GUILayout.Button("Reload", EditorStyles.toolbarButton, GUILayout.Width(60)) && (!Dirty || EditorUtility.DisplayDialog("Reload", "Discard unsaved changes?", "Discard", "Cancel"))) Load();`
`NumberMergeLevelEditor.cs:114`	`if (GUILayout.Button("Validate", EditorStyles.toolbarButton, GUILayout.Width(70)) && Cur != null) ValidateOne(_sel);`
`NumberMergeLevelEditor.cs:115`	`if (GUILayout.Button("Validate All", EditorStyles.toolbarButton, GUILayout.Width(85))) ValidateAll();`

NumberMergeLevelEditor.cs:118	`if (GUILayout.Button(new GUIContent("Play this level", "Optional quick test: saves, opens the WebGL scene at this stage and enters Play mode. Editing never needs it."), EditorStyles.toolbarButton, GUILayout.Width(100)) && Cur != null) Play(_sel + 1);`
NumberMergeLevelEditor.cs:146	`if (GUILayout.Button("Add"))`
NumberMergeLevelEditor.cs:151	`if (GUILayout.Button("Dup") && Cur != null) { var c = Cur.Clone(); c.name += " copy"; Insert(_sel + 1, c); }`
NumberMergeLevelEditor.cs:152	`if (GUILayout.Button("Del") && Cur != null && EditorUtility.DisplayDialog("Delete", $"Delete stage {_sel + 1} '{Cur.name}'?", "Delete", "Cancel"))`
NumberMergeLevelEditor.cs:156	`if (GUILayout.Button("Up") && _sel > 0) Move(-1);`
NumberMergeLevelEditor.cs:157	`if (GUILayout.Button("Down") && _sel < _file.levels.Length - 1) Move(1);`
NumberMergeLevelEditor.cs:186	`EditorGUILayout.LabelField($"Stage {_sel + 1}", EditorStyles.boldLabel);`
NumberMergeLevelEditor.cs:188	`l.name = EditorGUILayout.TextField("Name", l.name);`
NumberMergeLevelEditor.cs:190	`l.chapter = NmGen.ChapterNames[EditorGUILayout.Popup("Chapter", ci, NmGen.ChapterNames, GUILayout.Width(300))];`
NumberMergeLevelEditor.cs:192	`l.hint = EditorGUILayout.TextField(new GUIContent("First-time hint", "Shown on the stage card (introduce new things here; \\n for a line break)"), l.hint.Replace("\n", "\\n")).Replace("\\n", "\n");`
NumberMergeLevelEditor.cs:194	`int cols = EditorGUILayout.IntSlider("Jar width", l.cols, 4, 8);`
NumberMergeLevelEditor.cs:195	`int rows = EditorGUILayout.IntSlider("Jar height", l.rows, 5, 10);`

`NumberMergeLevelEditor.cs:199`	`l.target = EditorGUILayout.IntSlider(new GUIContent("Goal number", "Make this number to win"), l.target, 3, 40);`
`NumberMergeLevelEditor.cs:200`	`l.rise = EditorGUILayout.Slider(new GUIContent("Row timer (s)", "Seconds between rising rows; 0 = calm (a row rises only when no merge is in reach)"), l.rise, 0, 30);`
`NumberMergeLevelEditor.cs:203`	`l.spawnLo = EditorGUILayout.IntField(new GUIContent("Spawn low", "Smallest number in new rows"), l.spawnLo);`
`NumberMergeLevelEditor.cs:204`	`l.spawnHi = EditorGUILayout.IntField(new GUIContent("Spawn high", "Highest number in new rows at the start (it rises with the best sweet - gap)"), l.spawnHi);`
`NumberMergeLevelEditor.cs:205`	`l.spread = EditorGUILayout.IntSlider(new GUIContent("Spread", "How many different numbers a new row can hold"), l.spread, 1, 12);`
`NumberMergeLevelEditor.cs:206`	`l.gap = EditorGUILayout.IntSlider(new GUIContent("Gap", "New rows go up to best sweet - gap"), l.gap, 1, 6);`
`NumberMergeLevelEditor.cs:209`	`l.tieChance = EditorGUILayout.Slider("Tie chance/row", l.tieChance, 0, 1);`
`NumberMergeLevelEditor.cs:210`	`l.sugarChance = EditorGUILayout.Slider("Sugar chance/row", l.sugarChance, 0, 1);`
`NumberMergeLevelEditor.cs:211`	`l.iceChance = EditorGUILayout.Slider("Frozen chance/row", l.iceChance, 0, 1);`
`NumberMergeLevelEditor.cs:214`	`l.seed = EditorGUILayout.IntField(new GUIContent("Row seed", "Seeds the rising rows"), l.seed);`
`NumberMergeLevelEditor.cs:215`	`if (GUILayout.Button("New seed", GUILayout.Width(80))) l.seed = UnityEngine.Random.Range(1, 999999);`
`NumberMergeLevelEditor.cs:216`	`l.difficulty = EditorGUILayout.Slider("Design difficulty", l.difficulty, 0, 1);`
`NumberMergeLevelEditor.cs:217`	`l.breather = EditorGUILayout.ToggleLeft("Breather", l.breather, GUILayout.Width(80));`

NumberMergeLevelEditor.cs:219	l.layout = NmGen.Layouts[EditorGUILayout.Popup("Layout", li, NmGen.Layouts, GUILayout.Width(240))];
NumberMergeLevelEditor.cs:224	if (GUILayout.Button(new GUIContent("Regenerate start", "Rebuild the start layout from the layout style, goal and difficulty (new seed)"), GUILayout.Width(130))) { Regenerate(l); Changed(); }
NumberMergeLevelEditor.cs:225	if (GUILayout.Button(new GUIContent("Apply gravity", "Drop floating sweets and merge equal ones that rest on each other"), GUILayout.Width(110))) { ApplyGravity(l); Changed(); }
NumberMergeLevelEditor.cs:226	if (GUILayout.Button("Clear", GUILayout.Width(60)) && EditorUtility.DisplayDialog("Clear", "Empty the jar?", "Clear", "Cancel")) { l.cells = new int[l.cols * l.rows]; l.ties = new int[0]; Changed(); }
NumberMergeLevelEditor.cs:227	_showRoute = EditorGUILayout.ToggleLeft("Show bot's first drags", _showRoute, GUILayout.Width(160));
NumberMergeLevelEditor.cs:228	_showRows = EditorGUILayout.ToggleLeft("Preview next rows", _showRows, GUILayout.Width(140));
NumberMergeLevelEditor.cs:237	EditorGUILayout.LabelField($"Tiles {l.Tiles} start best {l.StartMax} goal {l.target} ties {l.ties.Length / 2} sugar {l.SugarCount} frozen {l.IceCount}" +
NumberMergeLevelEditor.cs:240	EditorGUILayout.HelpBox("Bot: " + v + $"\nA proof: the bot wins with one drag per {2.0f:0.0} s of row timer. Stars: 3 by row {(v.Solvable ? v.Rows + 1 : 0)}, 2 by row {(v.Solvable ? v.Rows + 4 : 0)}." +
NumberMergeLevelEditor.cs:258	if (_brush == Brush.Number || _brush == Brush.Frozen) _value = EditorGUILayout.IntSlider("Number", _value, 1, 40, GUILayout.Width(320));
NumberMergeLevelEditor.cs:260	EditorGUILayout.LabelField("Left-click/drag paints, right-click erases. Tie: click two side-by-side or stacked numbers (click a tied sweet to untie). Row 0 is the top (the overflow row).", EditorStyles.miniLabel);

NumberMergeLevelEditor.cs:475	EditorGUILayout.LabelField("Difficulty curve (bot score; blue = breather, red = not proven). Click a bar to select.", EditorStyles.miniBoldLabel);
NumberMergeLevelEditor.cs:497	_showGen = EditorGUILayout.Foldout(_showGen, "Auto Generate (seeded, bot-validated, same NmGen code as the runtime)", true);
NumberMergeLevelEditor.cs:500	_genCount = Mathf.Clamp(EditorGUILayout.IntField("Count", _genCount), 1, 100);
NumberMergeLevelEditor.cs:501	_genStart = Mathf.Max(1, EditorGUILayout.IntField("Start at stage #", _genStart));
NumberMergeLevelEditor.cs:502	_genSeed = EditorGUILayout.IntField("Seed", _genSeed);
NumberMergeLevelEditor.cs:505	_genFrom = EditorGUILayout.Slider("Start difficulty", _genFrom, 0, 1);
NumberMergeLevelEditor.cs:506	_genTo = EditorGUILayout.Slider("End difficulty", _genTo, 0, 1);
NumberMergeLevelEditor.cs:509	_genBreather = EditorGUILayout.IntSlider("Breather every", _genBreather, 0, 8);
NumberMergeLevelEditor.cs:510	_genReplace = EditorGUILayout.ToggleLeft("Replace that range (else insert)", _genReplace);
NumberMergeLevelEditor.cs:513	_genTies = EditorGUILayout.ToggleLeft("Ties", _genTies, GUILayout.Width(70));
NumberMergeLevelEditor.cs:514	_genSugar = EditorGUILayout.ToggleLeft("Sugar", _genSugar, GUILayout.Width(70));
NumberMergeLevelEditor.cs:515	_genIce = EditorGUILayout.ToggleLeft("Frozen", _genIce, GUILayout.Width(70));
NumberMergeLevelEditor.cs:526	if (GUILayout.Button($"Generate {_genCount} stage(s)", GUILayout.Height(26))) Generate();

CHAPTER 07

Generator, insertion and replacement

Use this editor generation panel, choose a reproducible seed and small count, review start/index and replace behavior, then generate, save and reopen. The source excerpt below identifies exact append/replace semantics.

Source	Exact current code
`NumberMergeLevelEditor.cs:40`	`bool _genReplace, _genTies = true, _genSugar = true, _genIce = true;`
`NumberMergeLevelEditor.cs:510`	`_genReplace = EditorGUILayout.ToggleLeft("Replace that range (else insert)", _genReplace);`
`NumberMergeLevelEditor.cs:556`	`if (_genReplace) { int rm = Mathf.Min(_genCount, list.Count - at); if (rm > 0) list.RemoveRange(at, rm); }`
`NumberMergeLevelEditor.cs:557`	`list.InsertRange(at, made);`

Worked extension: back up the bank, generate a small set at moderate difficulty using the exact per-game controls, validate them, then Save. Reopen the JSON and confirm the count changed. Do not assume every editor has the same insertion behavior: the actual implementation below distinguishes insert, append and replace. Runtime continuation does not automatically append new records to project JSON. Validation and bot scores help find problems but do not certify a pleasant human difficulty curve.

CHAPTER 08

Actual level JSON and identifiers

Canonical asset: `Assets/Games/NumberMerge/Resources/numbermerge/Levels/levels.json`. Array order controls visible numbering. Preserve numeric/boolean types, root version fields where present and resource/save identifiers. The actual first record is shown below; use the editor for structured coordinates and block/word data.


```
{
  "name": "First Taste",
  "hint": "Drag a sweet into the same number to merge them.\n2 + 2 makes a 3!",
  "chapter": "Gumdrop Lane",
  "layout": "flat",
  "cols": 5,
  "rows": 7,
  "target": 6,
  "rise": 0,
  "spawnLo": 1,
  "spawnHi": 3,
  "spread": 3,
  "gap": 2,
  "tieChance": 0,
  "sugarChance": 0,
  "iceChance": 0,
  "seed": 3031,
  "breather": false,
  "difficulty": 0,
  "cells": [
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
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    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    0,
    4,
    3,
    1,
    2,
    2,
    3,
    2,
    3,
    1,
    3
  ],
  "ties": [],
  "solvable": true,
  "par": 1,
  "botMoves": 5,
  "peak": 3,
  "score": 14.940477
}
```


Root keys: levels. Inspect NmCore.cs, NmGen.cs for field semantics and validation. Keep visible names/hints distinct from technical keys. The game-specific board, wave, card, opponent or question records are gameplay data; UI translation is

separate.

CHAPTER 09

Reskin recipe: interface colours

Stop Play mode and open `Scripts/NmArt.cs`. Find `Cream`, currently `#FFF4E0`. Change that hex literal to `#42C9B0`, preserve its symbol/type, compile and reopen the scene. Review HUD, menus, helper cards, settings/results and both orientations. Art sprites and procedural fallback shapes can have additional colours; edit them deliberately.

Actual consumers of this symbol are listed first, then all current palette/hex assignments. Editor-only palettes do not control the runtime. Preserve state/goal contrast and rebuild after accepting a change.

Source	Exact current code
`NmArt.cs:10 / class/static declarations`	`Cream = HzUI.Hex("#FFF4E0"), Choco = HzUI.Hex("#5A3220"), ChocoSoft = HzUI.Hex("#8A5A44"), Berry = HzUI.Hex("#C2457A"),`
`NumberMergeGame.cs:393 / public void OnBoardDown(Vector2 screen)`	`if (B.V[c] < 0) Toast("Sugar cubes can't move. Merge next to one to crack it!", Cream, 1.4f, ToastY);`
`NumberMergeGame.Hud.cs:619 / void ShowTip(string text)`	`_tipText = HzUI.Label("T", tip.transform, text, IsPortrait ? 38 : 34, Cream, TextAnchor.MiddleCenter, false);`

Source	Exact current code
`NmArt.cs:9 / class/static declarations`	`public static readonly Color Pink = HzUI.Hex("#FF8FB8"), Mint = HzUI.Hex("#4FCFA9"), Caramel = HzUI.Hex("#F2A65A"), Lemon = HzUI.Hex("#FFD84A"),`
`NmArt.cs:10 / class/static declarations`	`Cream = HzUI.Hex("#FFF4E0"), Choco = HzUI.Hex("#5A3220"), ChocoSoft = HzUI.Hex("#8A5A44"), Berry = HzUI.Hex("#C2457A"),`
`NmArt.cs:11 / class/static declarations`	`Sky = HzUI.Hex("#5AB8E8"), Grape = HzUI.Hex("#9B72D6"), Ice = HzUI.Hex("#BFEFFF"), Locked = HzUI.Hex("#C9AFA3"),`
`NmArt.cs:12 / class/static declarations`	`CardBg = HzUI.Hex("#FFF8EE"), PopupCard = HzUI.Hex("#FFF4E6"), PopupInner = HzUI.Hex("#FFE6EF"), Glass = HzUI.Hex("#FFFFFF"),`

NmArt.cs:13 / class/static declarations	JarBack = HzUI.Hex("#FBE3EC"), SlotCol = HzUI.Hex("#F6D2DF");
NmArt.cs:18 / class/static declarations	"#FF8FB8", // 1 strawberry
NmArt.cs:19 / class/static declarations	"#6FE0C0", // 2 mint
NmArt.cs:20 / class/static declarations	"#FFD84A", // 3 lemon
NmArt.cs:21 / class/static declarations	"#7CC6F2", // 4 blueberry
NmArt.cs:22 / class/static declarations	"#F2A65A", // 5 caramel
NmArt.cs:23 / class/static declarations	"#B48CF0", // 6 grape
NmArt.cs:24 / class/static declarations	"#A8E05A", // 7 lime
NmArt.cs:25 / class/static declarations	"#FF8A5C", // 8 orange
NmArt.cs:26 / class/static declarations	"#E8506E", // 9 cherry
NmArt.cs:27 / class/static declarations	"#4FC9C9", // 10 sour apple
NmArt.cs:28 / class/static declarations	"#FFB3A0", // 11 peach
NmArt.cs:29 / class/static declarations	"#8E9BFF", // 12 violet
NmArt.cs:50 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);	new Chapter { Name = "Gumdrop Lane", Tint = new Color(1f, 0.75f, 0.85f, 0.10f), Glow = HzUI.Hex("#FFE0EC") },
NmArt.cs:51 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);	new Chapter { Name = "Caramel Works", Tint = new Color(1f, 0.7f, 0.35f, 0.16f), Glow = HzUI.Hex("#FFE7C7") },
NmArt.cs:52 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);	new Chapter { Name = "Mint Mill", Tint = new Color(0.4f, 0.9f, 0.75f, 0.16f), Glow = HzUI.Hex("#D8FFF1") },
NmArt.cs:53 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);	new Chapter { Name = "Berry Vats", Tint = new Color(0.65f, 0.3f, 0.7f, 0.18f), Glow = HzUI.Hex("#F0DDFF") },
NmArt.cs:54 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);	new Chapter { Name = "Cocoa Tower", Tint = new Color(0.45f, 0.25f, 0.15f, 0.22f), Glow = HzUI.Hex("#FFE4D2") },

`NmArt.cs:55 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);`	`new Chapter { Name = "Endless", Tint = new Color(1f, 0.4f, 0.6f, 0.14f), Glow = HzUI.Hex("#FFE0EC") },`
`NumberMergeGame.cs:32 / class/static declarations`	`public static readonly string[] BoostColor = { "#FF6FA0", "#5AB8E8", "#B48CF0", "#F2A65A" };`
`NumberMergeGame.cs:354 / void ProcessEvents()`	`case NmEvent.Kind.Break: HzAudio.Play(Sfx.Break, 0.7f); BurstFx(e.Cell, HzUI.Hex("#FFFFFF"), 14); break;`
`NumberMergeGame.Hud.cs:309 / void RefreshTimerLook()`	`_timerFill.color = _chillT > 0 ? Sky : Calm ? HzUI.Hex("#BFE9DA") : Mint;`
`NumberMergeGame.View.cs:73 / void BuildView()`	`var g = HzUI.Img("BgImg", _bgLayer, HzUI.Gradient(HzUI.Hex("#FFE3EE"), HzUI.Hex("#F7B9D0")), Color.white); g.preserveAspect = false; g.raycastTarget = false;`
`NumberMergeGame.View.cs:221 / void ClearViews()`	`v.IceNum = HzUI.Label("IceNum", v.IceImg.transform, "", Mathf.Max(20, (int)(_tile * 0.52f)), HzUI.Hex("#2F5F7E"));`
`NumberMergeGame.View.cs:226 / void ClearViews()`	`if (!v.Tie.sprite) { v.Tie.sprite = HzUI.RoundRect(10); v.Tie.type = Image.Type.Sliced; v.Tie.color = HzUI.Hex("#C8283C"); }`
`NumberMergeGame.View.cs:247 / void Paint(TileView v, int i)`	`var tint = hp > 1 ? HzUI.Hex("#F6D2A6") : Color.white;`
`NumberMergeGame.View.cs:250 / void Paint(TileView v, int i)`	`else { v.Sugar.sprite = HzUI.RoundRect(14); v.Sugar.type = Image.Type.Sliced; v.Sugar.color = cracked ? HzUI.Hex("#E9DDD2") : tint; }`
`NumberMergeGame.View.cs:263 / void Paint(TileView v, int i)`	`v.Rim.color = val >= 30 ? HzUI.Hex("#FFFFFF") : Lemon;`

CHAPTER 10

Reskin recipe: visible menu and HUD text

Open `Scripts/NumberMergeGame.Hud.cs`, find the displayed literal `"Settings"`, and replace its displayed value with `"Options"`. Keep hierarchy names and technical IDs unchanged. In HzUI.Label/Button calls the first string may be an object name; change the displayed text argument. Read the method references below to distinguish construction from later refresh assignments.

Preserve interpolation braces, invariant number formatting, ON/OFF updates and line breaks. Update both initial and refresh paths, portrait and landscape. Exact helper names/unlocks/charges: Scripts/NumberMergeGame.cs; help and effect descriptions: Scripts/NumberMergeGame.Hud.cs. Level names/hints remain separate JSON content. If a title appears in art, Title property or another screen, update those visible uses deliberately without renaming GameId, namespace, resources or save keys.

Hard-coded game strings do not automatically use shared localization. To translate the interface, route every display assignment through your translation table. Dictionaries, word banks and theme pools are independent gameplay assets and need separate validation. Test help, settings, results, tutorial and helper introductions after a fresh launch.

Source	Exact current code
`NmArt.cs:50 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);`	`new Chapter { Name = "Gumdrop Lane", Tint = new Color(1f, 0.75f, 0.85f, 0.10f), Glow = HzUI.Hex("#FFE0EC") },`
`NmArt.cs:51 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);`	`new Chapter { Name = "Caramel Works", Tint = new Color(1f, 0.7f, 0.35f, 0.16f), Glow = HzUI.Hex("#FFE7C7") },`
`NmArt.cs:52 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);`	`new Chapter { Name = "Mint Mill", Tint = new Color(0.4f, 0.9f, 0.75f, 0.16f), Glow = HzUI.Hex("#D8FFF1") },`
`NmArt.cs:53 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);`	`new Chapter { Name = "Berry Vats", Tint = new Color(0.65f, 0.3f, 0.7f, 0.18f), Glow = HzUI.Hex("#F0DDFF") },`
`NmArt.cs:54 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);`	`new Chapter { Name = "Cocoa Tower", Tint = new Color(0.45f, 0.25f, 0.15f, 0.22f), Glow = HzUI.Hex("#FFE4D2") },`
`NmArt.cs:55 / public static Color Darker(Color c, float k = 0.25f) => Color.Lerp(c, new Color(0.25f, 0.08f, 0.1f), k);`	`new Chapter { Name = "Endless", Tint = new Color(1f, 0.4f, 0.6f, 0.14f), Glow = HzUI.Hex("#FFE0EC") },`
`NumberMergeGame.cs:29 / class/static declarations`	`public static readonly string[] BoostName = { "Pop", "Chill", "Sprinkle", "Scoop" };`

`NumberMergeGame.cs:135 / void ShowIntro()`	`NmPopup("Candy Rush unlocked!", "The classic endless jar: rows keep rising and the numbers follow your best sweet.\nHow far can you go? Find it under the Rush button.",`
`NumberMergeGame.cs:147 / void ShowLevelCard()`	`var p = NmPopup(LevelLabel(), body, Berry, CardArt(), (D.name == D.chapter ? D.name : D.name + " - " + D.chapter) + (D.breather ? " - breather" : ""), -1,`
`NumberMergeGame.cs:155 / string LevelLabel() => "Stage " + _playLevel.ToString(CultureInfo.InvariantCulture);`	`string LevelLabel() => "Stage " + _playLevel.ToString(CultureInfo.InvariantCulture);`
`NumberMergeGame.cs:206 / void Win()`	`NmPopup(stars == 3 ? "Delicious!" : "Batch Done!", msg, Mint, null, D.name, stars, null,`
`NumberMergeGame.cs:232 / void Lose()`	`NmPopup("Sweet Run!", $"Score {Fmt(_score)} Best {Fmt(BestScore)}\nBest sweet this run: {Fmt(B.Max)}", Berry, null, null, -1, null, buttons.ToArray());`
`NumberMergeGame.cs:237 / void Lose()`	`NmPopup("Jar Overflow!", "The jar is full to the brim.\nScoop the top two rows and keep going?", Berry, Get("booster_scoop"), null, -1, null, buttons.ToArray());`
`NumberMergeGame.cs:324 / void CheckAutoRise()`	`Toast(Calm ? "No merges in reach - here comes a row!" : "No pairs left - here comes a row!", Lemon, 1.1f, ToastY);`
`NumberMergeGame.cs:335 / void CheckMilestone()`	`Toast(m == 30 ? "You made a 30! Legendary!" : "Sweet " + Fmt(m) + "!", Lemon, 1.5f, ToastY);`
`NumberMergeGame.cs:393 / public void OnBoardDown(Vector2 screen)`	`if (B.V[c] < 0) Toast("Sugar cubes can't move. Merge next to one to crack it!", Cream, 1.4f, ToastY);`
`NumberMergeGame.cs:394 / public void OnBoardDown(Vector2 screen)`	`else Toast("Frozen! Merge into it or next to it to melt the ice.", Ice, 1.4f, ToastY);`
`NumberMergeGame.cs:493 / void OnBooster(int b)`	`Toast(BoostName[b] + " unlocks at stage " + Fmt(BoostUnlock[b]), null, 1.3f, ToastY); HzAudio.Play(Sfx.Cancel, 0.4f);`
`NumberMergeGame.cs:511 / void OnBooster(int b)`	`if (Calm) { Toast("No timer in this stage", null, 1.1f, ToastY); return; }`

`NumberMergeGame.cs:516 / void OnBooster(int b)`	`Toast("Row timer frozen for 15 s!", Ice, 1.3f, ToastY);`
`NumberMergeGame.cs:522 / void OnBooster(int b)`	`if (!any) { Toast("The bottom row is empty", null, 1.1f, ToastY); return; }`
`NumberMergeGame.cs:546 / void UseAim(int c)`	`if (B.V[c] <= 0) { HzAudio.Play(Sfx.Cancel, 0.4f); Toast("Sprinkle works on numbers", null, 1.1f, ToastY); return; }`
`NumberMergeGame.cs:576 / void OfferCharge(int b)`	`NmPopup("+1 " + BoostName[b], "Out of " + BoostName[b] + ".\nWatch a short ad for one more?", HzUI.Hex(BoostColor[b]), Get(BoostArt[b]), null, -1, null,`
`NumberMergeGame.cs:716 / string PopupTitle()`	`return t ? t.GetComponent<Text>().text.Replace(' ', '_') : "popup";`
`NumberMergeGame.Hud.cs:68 / void BuildHudLandscape(float W, float H)`	`var title = HzUI.Label("Title", _layer, "Number\nMerge", 74, Pink, TextAnchor.MiddleLeft);`
`NumberMergeGame.Hud.cs:78 / void BuildHudLandscape(float W, float H)`	`var rb = HzUI.Button("Restart", _layer, "", Caramel, Color.white, () => { if (_playing || _ended) Restart(); }, 36, 34);`
`NumberMergeGame.Hud.cs:80 / void BuildHudLandscape(float W, float H)`	`ButtonIconLabel(rb, HzIcons.Replay, "Retry");`
`NumberMergeGame.Hud.cs:81 / void BuildHudLandscape(float W, float H)`	`var sb = HzUI.Button("Settings", _layer, "", Grape, Color.white, ShowSettingsPopup, 36, 34);`
`NumberMergeGame.Hud.cs:83 / void BuildHudLandscape(float W, float H)`	`ButtonIconLabel(sb, HzIcons.Settings, "Menu");`
`NumberMergeGame.Hud.cs:84 / void BuildHudLandscape(float W, float H)`	`var rush = HzUI.Button("Rush", _layer, "", EndlessOpen || _endless ? Pink : Locked, Color.white, OnRushButton, 36, 34);`
`NumberMergeGame.Hud.cs:86 / void BuildHudLandscape(float W, float H)`	`ButtonIconLabel(rush, _endless ? HzIcons.Flag : EndlessOpen ? HzIcons.FastForward : HzIcons.Lock, _endless ? "Back" : "Rush");`
`NumberMergeGame.Hud.cs:94 / void LevelPill(Image pill)`	`_levelTxt = HzUI.Label("Level", pill.transform, "", 60, Color.white); _levelTxt.AddOutline(new Color(0.3f, 0.02f, 0.1f, 0.45f), 3);`

NumberMergeGame.Hud.cs:96 / void LevelPill(Image pill)	`_nameTxt = HzUI.Label("Name", pill.transform, "", 32, new Color(1, 1, 1, 0.96f), TextAnchor.MiddleCenter, false); _nameTxt.fontStyle = FontStyle.Bold;`
NumberMergeGame.Hud.cs:106 / void ButtonIconLabel(Button b, string icon, string label)	`var t = HzUI.Label("L", b.transform, label, 36, Color.white); t.rectTransform.Stretch(); t.rectTransform.offsetMin = new Vector2(62, 0);`
NumberMergeGame.Hud.cs:131 / void ButtonIconLabel(Button b, string icon, string label)	`num = HzUI.Label("Num", body.transform, Fmt(value), (int)(size * (value >= 10 ? 0.44f : 0.54f)), Choco);`
NumberMergeGame.Hud.cs:151 / void GoalCard(Image card, float w, float h)	`_goalTxt = HzUI.Label("Goal", card.transform, "", 40, Choco, TextAnchor.MiddleLeft);`
NumberMergeGame.Hud.cs:153 / void GoalCard(Image card, float w, float h)	`_goalSub = HzUI.Label("Sub", card.transform, "", IsPortrait ? 32 : 24, ChocoSoft, TextAnchor.MiddleLeft, false); _goalSub.fontStyle = FontStyle.Bold;`
NumberMergeGame.Hud.cs:166 / void GoalCard(Image card, float w, float h)	`var bl = HzUI.Label("BestLbl", card.transform, "best", 24, ChocoSoft, TextAnchor.MiddleCenter, false); bl.fontStyle = FontStyle.Bold;`
NumberMergeGame.Hud.cs:183 / void TimerBar(float cx, float cy, float w, float h)	`_timerTxt = HzUI.Label("TLabel", card.transform, "", IsPortrait ? 36 : 28, Choco, TextAnchor.MiddleLeft, false); _timerTxt.fontStyle = FontStyle.Bold;`
NumberMergeGame.Hud.cs:185 / void TimerBar(float cx, float cy, float w, float h)	`_rowsTxt = HzUI.Label("Rows", card.transform, "", IsPortrait ? 34 : 26, ChocoSoft, TextAnchor.MiddleRight, false); _rowsTxt.fontStyle = FontStyle.Bold;`
NumberMergeGame.Hud.cs:226 / void BoosterBar(float cx, float cy, float w, float h, bool horizontal)	`var btn = HzUI.Button("Booster" + b, parent, null, Color.white, Color.white, () => OnBooster(bb), 40, (int)(size * 0.32f));`
NumberMergeGame.Hud.cs:249 / void BoosterBar(float cx, float cy, float w, float h, bool horizontal)	`v.BadgeText = HzUI.Label("N", bd.transform, "", (int)(size * 0.24f), Color.white); v.BadgeText.rectTransform.Stretch();`
NumberMergeGame.Hud.cs:251 / void BoosterBar(float cx, float cy, float w, float h, bool horizontal)	`v.Caption = HzUI.Label("Cap", v.Root, "", 30, Choco);`
NumberMergeGame.Hud.cs:262 / void RefreshHud()	`_levelTxt.text = "CANDY RUSH";`

NumberMergeGame.Hud.cs:263 / void RefreshHud()	_nameTxt.text = "Endless - started from " + Fmt(_startFrom);
NumberMergeGame.Hud.cs:264 / void RefreshHud()	_goalTxt.text = "Score " + Fmt(_score);
NumberMergeGame.Hud.cs:265 / void RefreshHud()	_goalSub.text = "Best score " + Fmt(Mathf.Max(BestScore, _score));
NumberMergeGame.Hud.cs:271 / void RefreshHud()	_nameTxt.text = D.name == D.chapter ? D.name : D.name + " - " + D.chapter;
NumberMergeGame.Hud.cs:272 / void RefreshHud()	_goalTxt.text = "Make " + AOrAn(D.target) + "!";
NumberMergeGame.Hud.cs:273 / void RefreshHud()	_goalSub.text = "3 stars by row " + Fmt(Par + 1);
NumberMergeGame.Hud.cs:281 / void RefreshHud()	if (_rowsTxt) _rowsTxt.text = "rows " + Fmt(B.Rows);
NumberMergeGame.Hud.cs:302 / void UpdateTimerBar()	_timerTxt.text = sec == -2 ? "Calm: no timer" : sec <= -3 ? "Frozen " + Fmt(-3 - sec) + "s" : "Next row " + Fmt(sec) + "s";
NumberMergeGame.Hud.cs:326 / void RefreshBoosters()	v.BadgeText.text = c > 0 ? Fmt(c) : "+";
NumberMergeGame.Hud.cs:328 / void RefreshBoosters()	v.Caption.text = un ? BoostName[b] : "Stage " + Fmt(BoostUnlock[b]);
NumberMergeGame.Hud.cs:399 / HzPopup NmPopup(string title, string message, Color accent, Sprite art, string subtitle, int stars, Action<RectTransform> decorate, params (string label, Color color, Action action)[] buttons)	var t = HzUI.Label("Title", ribbon.transform, title, 70, Color.white); t.rectTransform.Stretch(6);
NumberMergeGame.Hud.cs:404 / HzPopup NmPopup(string title, string message, Color accent, Sprite art, string subtitle, int stars, Action<RectTransform> decorate, params (string label, Color color, Action action)[] buttons)	var st = HzUI.Label("Sub", card.transform, subtitle, 38, Berry); st.rectTransform.Place(new Vector2(0.5f, 1), new Vector2(cw - 80, 50), new Vector2(0, y - 25));
NumberMergeGame.Hud.cs:409 / HzPopup NmPopup(string title, string message, Color accent, Sprite art, string subtitle, int stars, Action<RectTransform> decorate, params (string label, Color color, Action action)[] buttons)	p.Message = HzUI.Label("Msg", card.transform, message, 42, Choco, TextAnchor.MiddleCenter, false);
NumberMergeGame.Hud.cs:439 / void GoalBadge(RectTransform area, string goal)	var t = HzUI.Label("Goal", area, goal, 66, Pink, TextAnchor.MiddleLeft);

`NumberMergeGame.Hud.cs:455 / void UnlockPopup(int b, Action done)`	`NmPopup("New: " + BoostName[b] + "!", BoostDesc[b] + "\nYou get " + Fmt(BoostStart[b]) + " free!", HzUI.Hex(BoostColor[b]), Get(BoostArt[b]), null, -1, null,`
`NumberMergeGame.Hud.cs:467 / void ShowSettingsPopup()`	`("How to Play", Caramel, () => HelpPage(0)),`
`NumberMergeGame.Hud.cs:473 / void ShowSettingsPopup()`	`var p = NmPopup("Menu", "", Grape, null, null, -1, null, list.ToArray());`
`NumberMergeGame.Hud.cs:482 / static string SoundLabel() => HzAudio.Muted ? "Sound: OFF" : "Sound: ON";`	`static string SoundLabel() => HzAudio.Muted ? "Sound: OFF" : "Sound: ON";`
`NumberMergeGame.Hud.cs:483 / static string MusicLabel() => HzAudio.MusicMuted ? "Music: OFF" : "Music: ON";`	`static string MusicLabel() => HzAudio.MusicMuted ? "Music: OFF" : "Music: ON";`
`NumberMergeGame.Hud.cs:498 / void HelpPage(int i)`	`NmPopup($"How to Play {i + 1}/{HelpPages.Length}", HelpPages[i], Sky, i == 0 ? Get("keyart") : i == 4 ? Get("sugar_cube") : null, null, -1, null,`
`NumberMergeGame.Hud.cs:499 / void HelpPage(int i)`	`(last ? "Got it" : "Next", Mint, last ? (Action)null : () => HelpPage(i + 1)));`
`NumberMergeGame.Hud.cs:508 / void OnRushButton()`	`Toast("Candy Rush opens after stage " + Fmt(EndlessUnlock - 1), null, 1.3f, ToastY);`
`NumberMergeGame.Hud.cs:521 / void ShowRushPopup()`	`NmPopup("Candy Rush", "The endless jar. Rows speed up as your numbers grow.\nBest score " + Fmt(BestScore) + " - best sweet " + Fmt(BestNumber), Pink, Get("keyart"), null, -1, null, list.ToArray());`
`NumberMergeGame.Hud.cs:619 / void ShowTip(string text)`	`_tipText = HzUI.Label("T", tip.transform, text, IsPortrait ? 38 : 34, Cream, TextAnchor.MiddleCenter, false);`
`NumberMergeGame.Hud.cs:707 / void CompleteGuide(int b)`	`if (b == BPop || b == BSprinkle) Toast("Nice! " + BoostName[b] + " is yours", Lemon, 1.3f, ToastY);`
`NumberMergeGame.View.cs:212 / void ClearViews()`	`v.Num = HzUI.Label("Num", v.Body.transform, "", Mathf.Max(20, (int)(_tile * 0.52f)), Choco);`
`NumberMergeGame.View.cs:221 / void ClearViews()`	`v.IceNum = HzUI.Label("IceNum", v.IceImg.transform, "", Mathf.Max(20, (int)(_tile * 0.52f)), HzUI.Hex("#2F5F7E"));`

`NumberMergeGame.View.cs:431 / void ComboPop(int n)`	`var t = HzUI.Label("Combo", _fxLayer, "Sweet x" + Fmt(n) + "!", IsPortrait ? 84 : 70, Lemon);`

CHAPTER 11

Artwork, fonts, audio and licences

Replace game images under Resources/numbermerge through Unity while retaining resource keys, .meta GUIDs, Sprite/UI import settings, alpha, pivots and slicing. Procedural graphics in the art/view sources may need separate colour changes. Replacing one PNG does not change every fallback. Keep third-party font and other licence notices with the corresponding shipped assets.

Effects are under Resources/numbermerge/Sfx. Preserve each actual load key and importer settings when replacing sounds. Supplied music: Resources/numbermerge/Music/candy_loop.ogg. Preserve that load key and verify loop/compression and the independent Music toggle. Kit UI uses supplied fonts; preserve actual licence notices and verify new glyphs.

CHAPTER 12

Local saves, reset and migration

HzSave wraps PlayerPrefs and flushes local state on focus loss/quit. WebGL persistence uses browser storage tied to the hosting origin. Clearing site data or changing origin starts a new local save; this is not cloud synchronization. Inspect HzSave/PlayerPrefs calls in the actual game source before altering progression, stars, helper charges or tutorial flags.

Verify fresh launch, a completed arena, helper use, sound options, focus loss and restart. Use a dedicated test profile for resets. Keep technical save keys stable during colour/text reskins. If changing schema or keys, implement and test a migration rather than silently discarding existing users' progress.

CHAPTER 13

Advertising integration and configuration

Asset: `Assets/Szekipapa77Ads/Resources/AdsSettings.asset`. Inspect backend, useGoogleTestIds and missingSdkFallback in the publisher-owned Ads configuration. Buyer projects include adapter/integration code and exclude external provider SDK binaries. WebGL and a missing-provider fallback simulate ads; these are not live commercial impressions.

For real Android ads install the supported provider SDK from its official source into your working project, enter your own app/placement IDs and use test ads first. Verify banner/interstitial/rewarded callbacks on supported hardware. For production use your own IDs and disable simulated missing-SDK fallback. Build tooling rejects unsafe production fallback settings. No IAP, remote backend or HLK is included in this publisher edition.

CHAPTER 14

Android build and release

Install Unity 6000.0.62f1 Android modules through Hub. Select Android, run Setup and review application ID `com.szekipapa77.numbermerge`, version, scene orientation and signing. The build command writes `Builds/Android/numbermerge.apk`. Confirm BuildResult.Succeeded and the actual output file; intermediate report totals are not the final APK size.

Use your own keystore/signing and distribution settings. A test-signed demo is not a signed store release. A provider-free build simulates ads; a provider test-demo does not prove production ad-account configuration. Test real hardware touch, safe areas, focus/audio, save/relaunch and reward callbacks. Keep passwords, keystores and private service credentials out of buyer packages.

CHAPTER 15

WebGL build and hosting

Install matching WebGL support. Run Setup and Build WebGL; output is `Builds/WebGL/numbermerge/`. Serve index.html with all loader/framework/data/WASM/StreamingAssets files over HTTP(S), not file://. Match compression, Content-Encoding and MIME types to the host. Outer page sizing and actual scene orientation are different settings.

The publisher's playable demo is hosted on HyperCodeStore. Upload the actual runtime files rather than merely changing a URL. Test a fresh anonymous load, real pointer controls, audio after interaction, saved progress/reload and phone/desktop sizing. Rebuild and rehost when code/assets change. A successful HTTP page alone is not gameplay verification.

CHAPTER 16

Troubleshooting

Compile errors: restore matching packages and the full private Kit/Ads layer; only one game Kit belongs in a project. **Pink/invisible rendering:** run Setup and inspect Graphics and every Quality URP assignment. **No input:** classic input handling must be enabled; check canvas focus and blocking modals. **Missing resources:** restore exact paths, .meta GUIDs and importer settings.

Edits do not appear: stop Play, Save in the level editor, reload the canonical bank and check current saved level. **Text reverts:** update construction and refresh assignments from the exact table. **No live ads:** provider binaries are deliberately excluded; inspect your installed provider and fallback mode. **Scene reload exception:** retain owner-lifetime and generation guards on delayed callbacks. **WebGL fails:** inspect browser Console, compression headers and missing payloads.

CHAPTER 17

Support and final release checklist

Publisher: Szekipapa77. Support: szekipapa77@gmail.com. Include game title, exact Unity version, platform, first Console error and reproduction steps. Never send passwords/tokens.

1. Clean-import the standalone project and native package.
2. Play both scenes and replace a scene near startup.
3. Review tutorial, normal completion/retry and every helper activation plus effect.
4. Validate editor changes and test reskin text/colours in both layouts.
5. Rebuild affected Android/WebGL outputs.
6. Verify complete public APK bytes, actual browser demo, documentation and video links.
7. Keep licences and exclude all internal prompts, planning/capture/build/QA tooling from buyers.